



Misrepresentation of the risk of ovarian cancer among women using menopausal hormones. Spurious findings in a meta-analysis

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ABSTRACT

Background: Based on a meta-analysis of 52 studies, and principally on a meta-analysis of 17 follow-up studies, it has been claimed that current-or-recent use (last use <5 years previously) of menopausal hormones causes ovarian cancer, even if the duration of use was <5 years, and that women aged about 50 years who use hormones for >5 years have about one extra case per 1000 users, and one extra fatal case per 1700 users.

Objective: To evaluate the validity of the evidence.

Methods: Generally accepted epidemiological principles of causation were applied to the evidence.

Findings: The study base included hysterectomised women, an unknown proportion of whom were oophorectomised, and not at risk for ovarian cancer. The findings did not satisfy the criteria of time order, bias, confounding, strength of association, dose–response, duration–response, consistency, and biological plausibility.

Conclusions: The meta-analysis did not establish that current-or-recent use of menopausal hormones causes ovarian cancer. The strong likelihood is that early symptoms of as yet undiagnosed ovarian cancer “caused” current-or-recent short-duration hormone use, not the reverse. The representation of the number of extra cases, and fatal cases, among hormone users was misleading and alarmist.

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1. Introduction

Beral and her colleagues, writing on behalf of the Collaborative Group on Epidemiological Studies of Ovarian Cancer, have recently reported on the risk of ovarian cancer among users of menopausal hormone therapy (MHT) in a meta-analysis of 52 studies [1,2]. The principal analyses, however, were confined to 17 follow-up studies. The risks for ever-use of MHT, for current or recent use (stopped <5 years before diagnosis), and for durations of current-or-recent use of <5 years and ≥5 years were significantly increased. Risks for current-or-recent use were increased for serous and endometrioid tumours, but not for mucinous or clear cell tumours.

The investigators concluded that “the increased risk may well be largely or wholly causal”.

Below we evaluate the validity of the evidence.

2. Summary of the findings

In the 52 studies there were 21 488 cases, and in the 17 follow-up studies there were 6601 cases; the respective numbers of cases exposed to MHT were 9303 (43.3%) and 6601 (54.5%). To minimize the likelihood of bias, the principal analyses were confined to the follow-up studies. In sensitivity analyses, however, the combined data in all 52 studies were evaluated.

In the follow-up studies up to four randomly selected controls were matched to each case, and individual data were analysed as case-control comparisons. In a Danish study [3], because of data protection laws, individual data were not available; instead, tabulated data were provided by the Danish investigators, and incorporated into the analyses.

With never-use of MHT as the reference category, relative risk (RR) estimates were as follows: ever-use, 1.20 (99% confidence interval (CI) 1.13–1.28); current-or-recent use for durations of <5 years and >5 years, 1.43 (95% CI 1.31–1.56) and 1.37 (95% CI 1.29–1.43), respectively. Among women who used MHT for >5 years the risk declined after stopping, but was still elevated 10 years later

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(RR 1.10; 95% CI 1.01–1.20). Among current-or-recent MHT users the RRs for serous and endometrioid tumours were 1.53 (95% CI 1.40–1.66) and 1.42 (95% CI 1.20–1.67), respectively; for mucinous and clear cell tumours the RRs were 0.93 and 0.75. A test for heterogeneity among the four histological subtypes was significant ($p < 0.0001$).

Under causal assumptions, the investigators estimated that “women who use [MHT] for 5 years around age 50 years have about one extra case per 1000 users and . . . about one extra ovarian cancer death per 1700 users”.

3. Evaluation

Below, we apply generally accepted epidemiological criteria of causation [4,5] to the evidence from the meta-analysis. The criteria are inter-related, and when appropriate, we cross-refer. Before doing so, however, since the principal analyses were confined to the 17 follow-up studies, it is first necessary to mention that 49.7% (6022/12 110) of the cases were derived from the Million Women Study (MWS) [6] (our calculation; appendix [2] p. 10). One study only provided data for 1258 fatal cases [7]; among the remaining 16 studies the MWS cases accounted for 55.5% (6022/10 852) of all cases. Thus the overall findings in the follow-up studies were mainly driven by a single study, the MWS.

4. Time order

The strong likelihood is that as yet undiagnosed ovarian cancer “caused” current-or-recent MHT use for <5 years, not the reverse, and time order was violated. That violation was by far the most credible explanation, if not the entire explanation, of the increased risk of ovarian cancer associated with short-duration use. Common presenting symptoms of ovarian cancer include discomfort or pain during sexual intercourse, lower abdominal discomfort or pain, abdominal distension, urinary difficulties, and recurrent bladder infections. Each of these symptoms could selectively have resulted in the use of MHT before a diagnosis of ovarian cancer was made, because they were initially attributed to the menopause.

With the exception of the mortality study [7], it is likely that time order was violated in all the remaining 16 studies.

There is no experimental evidence to indicate that short-duration MHT induces ovarian carcinogenesis (see Section 12), further supporting the likelihood of violation of time order.

5. Specification of the study base

In the follow-up studies, among current-or-recent MHT users information on hysterectomy status was provided for only 34.3% (737/2151) of the exposed cases (our calculation; appendix p. 15). Among the 737 exposed cases, 550 (74.6%) had undergone hysterectomy.

Hysterectomised women who had their ovaries removed at the same time would commonly have been unaware of it, and the proportion of women in the study base who were not at risk for ovarian cancer was unknown. The investigators speculated that the inclusion of oophorectomised women would have resulted in underestimation of the RR [1,8]. That speculation is not defensible. Hysterectomised women who retained one or both ovaries, and who developed symptoms of as yet undiagnosed ovarian cancer, would selectively have been current-or-recent users of MHT (see Section 4). Among hysterectomised women who retained their ovaries it is also likely that on prolonged follow-up ovarian cancer would have been diagnosed earlier among MHT users than among non-users (see Section 6).

In the absence of information on hysterectomy status for 73.3% of current-or-recent users of MHT, coupled with likelihood that the majority had undergone hysterectomy, the findings in the meta-analysis were uninterpretable (see Section 7).

6. Bias

Among 2671 current-or-recent MHT-using cases 277 (10.4%) were borderline (our calculation; appendix p. 19).

Borderline tumours would selectively have been diagnosed among MHT users, because they would have undergone vaginal examinations more frequently than non-users, when they renewed their prescriptions. That bias would have occurred for all durations of use, and it would have occurred in all but the single mortality study [7]. For the same reason, malignant tumours would have been diagnosed earlier among MHT users than among non-users.

Follow-up rates among the 17 studies were not mentioned in the report or in the appendix. However, in a previous report from the MWS [6], the rate was 64%, and with the possible exception of the Danish study [3] it is likely that substantial proportions of MHT users were lost to follow-up. Users who developed ovarian cancer would less commonly have been lost to follow-up than users who did not (see the following section).

7. Confounding

As mentioned above, losses to follow-up were substantial, and women who developed symptoms of as yet undiagnosed ovarian cancer would selectively have been prescribed MHT (see Section 4), and selectively have been followed (see Section 6). Women who used MHT for >5 years would selectively have been followed after they stopped.

Also as mentioned above, among current-or-recent MHT users information on hysterectomy status was provided for only 34.3% of the cases (see Section 5), among whom 74.6% had undergone hysterectomy. Among the remaining cases with unknown hysterectomy status it is likely that a substantial proportion also had hysterectomies. With missing data for 65.7% of the cases, adjustment for confounding by hysterectomy status was not possible.

The investigators argued that the heterogeneity in the RRs for the four histological subtypes of ovarian cancer “argues strongly for causality, because it implies that the [MHT]-associated risks were not due just to confounding and that different ovarian cancer types have different causes.”

That implication is not defensible [8,9]. First, it is not established that different subtypes of ovarian cancer have different causes. Second, histology can vary at different sites in the same tumour [9,10]. Third, among pathologists there is major discordance in diagnosing histological subtypes [9]. Fourth, in different studies different classifications of ovarian malignancies have been used [11]. For these reasons, precise and replicable diagnoses can only be made by pathologists with special training and experience in gynaecological pathology [9,11]. These considerations would have applied with added force to the meta-analysis, in which the quality of the pathological evidence across the studies would have varied.

In the absence of central and “blinded” audit, the histological classification presented in the meta-analysis was uninterpretable. And if, for the moment, that uninterpretability is set aside, since estrogen therapy causes endometrial cancer, a diagnosis of endometrioid ovarian cancer could selectively have been made by pathologists made aware of MHT use (see Section 6).

8. Strength of association

All summary RRs estimated in the meta-analysis were 1.53 or less. For such low-magnitude associations it is impossible to discriminate among bias, confounding, and causation as alternative explanations [12], and the combined effects of improper specification of the study base, violation of time order, bias, and confounding, support the likelihood that the associations were not causal.

In sensitivity analyses the data were truncated after variable durations of follow-up, and the findings were not materially changed. However, there was no *a priori* reason why they should have been. Sensitivity analyses to evaluate how much violation of time order, or how much bias or confounding it would have taken to account for the findings were not carried out, and minimal distortion could have accounted for the associations.

9. Statistical stability

Numbers in the meta-analysis were massive. In the follow-up studies, with 2751 of 6001 (41.7%) cases currently-or-recently exposed to MHT, virtually any RR that deviated from 1.00 would have been “highly significant”. Consequently, all associations identified in the meta-analysis were statistically stable, whether or not they were causal or due to bias or confounding.

10. Dose/duration–response

Dose-response was not evaluated, and under causal assumptions a gradient of increasing risk with increasing dose would have been expected. Duration-response was evaluated, and an increasing risk with increasing duration of use would also have been expected. Yet the respective RRs for current-or-recent use of MHT for <5 years and >5 years were 1.43 and 1.37. The absence of a duration-response effect was further evidence to support the likelihood that the associations were not causal.

11. Consistency

The investigators acknowledged that “the RR for current-or-recent [MHT] use did differ significantly between prospective and retrospective studies 1.37 vs 1.04; heterogeneity $p < 0.001$). . . due to the lack of an apparent effect in the aggregated North American retrospective studies”. That is, the findings across the 52 studies were discordant. However, they argued that the lack of an effect in the American case-control studies may have occurred because “the design might have made unbiased recruitment of controls difficult, and the average year of diagnosis was 1990, well before [MHT] had become common”.

That argument was not defensible. First, contrary to what was claimed, in 1990 among American women the use of MHT was already common enough for the case-control studies to be robust: some 10% of American women used MHT [13]; thereafter the use of MHT became more common. Second, irrespective of how commonly MHT was used, there is no reason why the timing of the case-control studies would have biased the selection of the controls. Third, the case-control studies were questionnaire- or interview-based. If anything, biased recall of MHT use would therefore have occurred in the case-control studies, but not in the follow-up studies (see Section 6). That bias would have given rise to higher RR estimates in than in the follow-up studies. Yet the reverse was the case. Fourth, the inconsistency of the findings in the European and American case-control studies was unexplained.

In the combined analyses of the follow-up and case-control data the findings were not materially changed. However, combining the

data simply served to camouflage the inconsistencies across the 52 studies included in the meta-analysis. Those inconsistencies further support the likelihood that the associations were not causal.

12. Biological plausibility

Experimental research on the possible carcinogenic effects of MHT on the ovary is sparse. The ovaries contain receptors which could mediate MHT-proliferating effects [14], and some proliferation cannot be excluded. However, compared with other proliferating effects, such as stromal growth factors, preliminary experimental research has shown that hormonal effects are not important [14]. In addition, there is no experimental evidence to suggest that exposure to estrogen, or to estrogen combined with progestogen, induces ovarian carcinogenesis.

13. Conclusions

Contrary to what has been claimed by the investigators, the combined effects of improper specification of the study base, violation of time order, bias, confounding, low-magnitude RRs, absence of information on dose-response, absence of a duration-response effect, inconsistency across the studies, and lack of biological plausibility, strongly suggest that the associations identified in the meta-analysis were not causal. By far the most plausible explanation for the association of short-duration current-or-recent MHT use with ovarian cancer is that early symptoms of as yet undiagnosed ovarian cancer “caused” hormone use, not the reverse.

Despite the multiple defects in the meta-analysis, if for the sake of the argument the findings are nevertheless assumed to be valid, what are the public health implications? The investigators stated that “the increased risk may well be largely or wholly causal; if it is, women who use [MHT] for 5 years around age 50 years have about one extra ovarian cancer per 1000 users, and . . . about one extra ovarian cancer death per 1700 users”; for 10 years of use the corresponding estimates were 1 in 600 and 1 in 800 users (appendix p. 11).

Those estimates were misleading and alarmist. Absolute risks are properly estimated as rates per year, not as rates per 5 or 10 years. Under causal assumptions, among women who used MHT for 5 years at age 50 years the excess incidence of ovarian cancer would thus have been 0.2 extra cases *per 1000 per year* (our calculation; appendix p 11), and the excess mortality would have been 0.12 extra cases *per year*; for 10 years of use at age 50 years the corresponding estimates would have been 0.18 and 0.12. Such exceptionally low absolute risks would have no significant public health implications.

Finally, the investigators stated that “the definite risk of ovarian cancer. . . even with less than 5 years of use. . . is directly relevant to medical advice, personal choices, and current efforts to revise UK and worldwide guidelines”. That assertion, particularly the claim that the risk was definite, was unjustified and alarmist.

Conflict of interest statement

The authors presently consult, and in the past have consulted, with manufacturers of menopausal hormones.

Contributors

Dr Shapiro conceived the paper and prepared the first draft.

Dr Stevenson, Dr Mueck and Dr Baber reviewed and edited the first draft and contributed to the final submission document.

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References

- [1] Collaborative Group on Epidemiological Studies. Menopausal hormone use and ovarian cancer risk: individual participant meta-analysis of 52 epidemiological studies. *Lancet* 2015, [http://dx.doi.org/10.1016/S0140-6736\(14\)61687-1](http://dx.doi.org/10.1016/S0140-6736(14)61687-1).
- [2] Collaborative Group on Epidemiological Studies. Menopausal hormone use and ovarian cancer risk: individual participant meta-analysis of 52 epidemiological studies. Supplementary appendix. *Lancet* 2015, [http://dx.doi.org/10.1016/S0140-6736\(14\)61687-1](http://dx.doi.org/10.1016/S0140-6736(14)61687-1).
- [3] Mørch LS, Lokkegaard E, Andreassen AH, et al. Hormone therapy and ovarian cancer. *JAMA* 2009;302:289–305.
- [4] Hill AB. The environment and disease: association or causation? *Proc R Soc Med* 1965;58:295–300.
- [5] Susser M. Causal thinking in the health sciences. New York, NY: Oxford University Press; 1979.
- [6] Beral V, Reeves G, Green J, Bull D, for the Million Women Study Collaborators. Ovarian cancer and hormone replacement therapy in the Million Women Study. *Lancet* 2007;369:1703–10.
- [7] Rodriguez C, Patel AV, Calle EE, Jacob EJ, Thun MJ. Estrogen replacement therapy and ovarian cancer mortality in a large prospective study of US women. *JAMA* 2001;285:1460–5.
- [8] Beral V, Green J, Reeves G. On behalf of the Million Women Study. Authors' response. *Climacteric* 2007;10:469–70.
- [9] Shapiro S. False alarm: postmenopausal hormone therapy and ovarian cancer. *Climacteric* 2007;10:466–70.
- [10] Kaufman DW, Kelly JP, Welch WR, et al. Noncontraceptive estrogen use and epithelial ovarian cancer. *Am J Epidemiol* 1989;130:1142–51.
- [11] Prat HJ. Pathology of the Ovary. Philadelphia, PA: Saunders; 2004.
- [12] Shapiro S. Bias in the evaluation of low-magnitude associations: an empirical perspective. *Am J Epidemiol* 2000;151:939–45.
- [13] Connelly MT, Richardson M, Platt R. Prevalence and duration of postmenopausal hormone replacement therapy in a managed care organization, 1990–1995. *J Gen Int Med* 2000;15:542–50.
- [14] Seeger H, Wallweiner D, Mueck AO. Is there a protective role of progestogens on the proliferation on human ovarian cancer cells in the presence of growth factors? *Eur J Gynec Oncol* 2005 (ISSN: 0392-2936).